

"Physics as Information Processing" ~ Chris Fields ~

Lecture 4

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foreign welcome everyone this is physics as information processing session 4 on communicating observers it's the 10th of August 2023 Chris looking forward to this lecture off to you uh thank you Daniel and uh welcome to this session uh as Daniel said this is about communicating observers and fastening your seat belts because we have a lot to cover in the next hour oh so just to do a quick review we've been talking about uh Quantum information Theory and we've been talking about it in the simplest possible setting of uh two systems Alice and Bob that exchange information through their Mutual boundary and we've assumed that everything is finite dimensional so we don't need to worry about infinite energy transfer across this boundary and of course we assume conservation of energy so there are no sources or sinks uh in either the systems Alice and Bob and we emphasize that this way of uh depicting things is topological we're not assuming any background geometry no sort of spatial embedding which is going to become very important in this session um in the last session we talked about how the boundary which I'll always call B it can be thought of as a channel that comprises in qubits that are independent and Alice and Bob each can prepare and measure those qubits and they can always choose independently their reference frames for doing so so how they Define up and down when they're measuring a spin with effectively a z-spin operator to to interact with these qubits we talked a little bit about the free energy principle in its Quantum formulation and the principle basically says that interacting systems will behave in a way that aligns their reference frames and that they therefore asymptotically approach entanglement so the fep is a classical limited the principle of unitarity and that's one reason for studying Quantum information to learn more about the fep and so to learn more about active inference and what the theory of active inference says about agents and last time we talked a lot about the constraints that are imposed by a limited free energy Supply on any agent uh recall that we talked about the fact that in the absence of sources of free energy uh within an agent so as long as you conserve energy globally that the free energy that each agent uses has to come from its

environment that's clearly true for us we have to consume oxygen and eat food and things like that to keep running as systems and we discussed a little bit this theorem that if two reference frames or Quantum processes q_1 and Q_2 don't commute they have to be implemented by compartments that communicate classically and we'll use this extensively today uh but the bottom line last time was that physics alone gives us compartmentalization in systems whenever the systems are complex enough to have multiple kinds of measurements that they can't deploy simultaneously so where we're going uh today we're going to start from this place we left off last time of the possibility of agents that are compartmentalized and into compartments that have to communicate classically so we're going to talk about agents that can be compartmentalized into multiple sub-agents that share an environment the environment consisting of Bob and how those agents talk about that shared environment so the motivation for this is simple it's that what we do all the time is a group activity and what we're doing right now is a group activity that involves sharing an environment what's shown on this screen and talking about it and the whole post session discussion process will be another example of just that sort of thing so to to talk about science at all or to talk about any kind of social relations or to talk about anything other than solipsism we have to be able to talk about how multiple agents communicate about some and so that's what we're going to develop a formal way of talking about and there are two things to notice right off the top about this one is that the way we've defined interactions of boundaries they're effectively instantaneous so each agent remember interacts with its boundary with some set of operators and what those operators do is effectively measure each operator effectively measures the spin of a qubit so this is a very fast process I'll the other thing that we then need to notice is that if we have two agents and they're communicating with each other classically then they're going to be using degrees of freedom of their shared environment in order to communicate and that's exactly what we're doing right now for example we're using degrees of freedom of the internet to communicate I'm using degrees of freedom in the local atmosphere to talk to my microphone and I'm using degrees of freedom of the local Photon field to see my screen so all communication between agents and certainly all classical communication between agents uses degrees of freedom of the environment and the environment remember is the other system Bob that's on the other side of that shared boundary so we're going to be interested in how components of Alice use Bob as a communication Channel and in fact as multiple communication channels and I just wanted to to quote my colleague and friend Mike Levin in his statement that all intelligence is composite all of us all organisms even single cells are composite systems that comprise many

many different agents that are able to make different kinds of measurements and execute different kinds of actions and so have to communicate classically so we're we're talking about all of life here as well as all of social systems and ecosystems and such things as that so the first thing I want to do today is simplify the notation uh so that we don't have to draw quite such complicated diagrams so the picture on the left here is a picture from the last session it shows the structure of a very simple agent that makes some measurements using a reference frame called e and then writes some results of those measurements to memory using a reference frame called y and those two processes are connected by an internal clock that ticks uh between the the input process and the output process and the the clock tick corresponds to the expenditure of free energy to write information irreversibly on the boundary so the the memories are actually written on the boundary because that's where classical information lives in this Theory so you can think of this whole process from E through the clock and back to Y as a single Quantum process or as a Quantum computation that Maps some sector e of the boundary to some sector y of the boundary and so we can represent it in this much simpler way that's over on the right side as some process Q that Maps some sector on the boundary to some other sector on the boundary and formally that process can be thought of as a topological Quantum field Theory uh topological here is important it's not geometric it doesn't assume an embedding in space and this will become critically important uh later on and the operation of that uh pqft requires a clock tick or it incorporates a tick of this internal clock and we can do this uh because we were able to prove Jim glasberg and and to Nino Marciano and I were able to approve last year that we can always represent a Quantum reference frame as a topological Quantum field Theory and this just illustrates the two kinds of Transformations between uh Quantum reference frames that are important we can take one and divide it into two pieces that commute or we can take uh a two-part reference frame and we can swap in uh another part that doesn't commute with the part that we've swapped it in for but that does commute with the remainder of the reference frame so here R is the reference component of some object identification system and we can swap in some different way of looking at the object once we've identified it so for example if the the object that we're identifying is a box that can measure both z -spin and X spin then p and Q can be the z -span and X spin settings on that box that allow us to to effectively rotate the filter from this way to that way so that we measure vertical spin versus for horizontal spin and that's really it I mean those are the the ways that one can transform uh what a reference frames of course we could do arbitrarily many compositions of this kind of transformation to do something very complicated but they map

very simply to tqfts with the structure um you can construct categories out of this and find a functor that's well defined at Mass one category to another so it's a it's an example of an issues of categorial categorical techniques for solving a problem that looks difficult but in fact it's it's very straightforward using this simplified notation um it becomes very easy to see uh that if we have two systems Alice and Bob on opposite sides of this boundary and we perfectly align their Quantum reference frames which is what the fep says they will tend to do then they become entangled and you can see that they're entangled by seeing that once the reference frames are perfectly aligned they jointly Implement one Quantum process um and jointly implementing one Quantum process just means you're entangled of the two sides no longer have um conditionally independent states if one Quantum process which is time reversible information preserving is implemented jointly by the two so a lot of things like this become very simple when we think of of these processes as just Quantum computations going on inside some system implemented by some system and how those processes relate across some boundary so that's what we're going to do for studying communication today another thing to recall from a couple of sessions ago or last session is that if two reference frames q_1 and Q_2 commute we can always join them together to form a single reference frame and this is just a graphical depiction in this simpler language of how that works topologically there's no distinction between the left side and the right side what I've done is just squeeze the top two green circles together and the bottom two green circles together into one Circle and that just makes a simple process connecting the top one to the bottom one so what we're going to be interested in is situations where this doesn't happen uh where Alice consists of two components that execute reference frames that don't commute and so the two components are conditionally independent they do each have free choice of reference frame and so they have to communicate classically they don't share a Quant a single Quantum process so they're not entangled and before we go ahead to do that I just wanted to remind you of some of the no-go theorems that follow from this situation uh depicted on the right where we have a single agent executing a single reference frame a single process a single kind of measurement uh any set of commuting measurements that can be combined into one measurement and here's some things Alice can't do and we've talked about some of these previously um so this is basically a reminder but I'll Alice can't using just one measurement measure the entanglement entropy across her own boundary so Alice can't know that she has an independent state she can't know that she's not entangled with Bob and that's that's a useful thing to think about philosophically uh as as it applies to you and me we can't actually know that we have independent states uh

Alice can't measure her own entanglement interview so she can't know whether she has components that have to communicate classically uh she can't measure Bob's entanglement entropy so she can't know whether Bob has compartments that have to and if you think in terms of perception Theory this says Alice can't know whether Bob consists of separate objects that behave classically oh their behavior can look classical to her but she's not able to determine that their behavior is classical because she can't uh measure Bob's entanglement entropy she can't measure the dimension of of and recall from a couple of times ago that the boundary includes some sector that supplies free energy that Alice can't extract information from for further processing she can only extract free energy from so she can't measure the dimension of that sector and so she can't measure the dimension of the whole boundary itself and this of course means that she can't measure her own Dimension either and of course she can't measure Bob's Dimension because the only information she can about get about Bob is the classical information that's encoded on this boundary if you can't reach past the boundary to see how large a system Bob is and that's clearly connected to this first point that you can't measure the entanglement entry across Bob because the entanglement attribute can only be small if her dimension in Bob's Dimension are both much larger than the dimension of the boundary so that's a condition she can't check um so all of these kind of no-go theorems lead off an interesting Direct s and the only one that we're really going to be interested in here is uh the third one the one about measuring Bob's entanglement entropy because we want to talk specifically about how multiple agents so multiple components of Alice can access information about Bob's entanglement okay so recall from the end of the last session that the last thing we talked about was classical communication and we talked about the fact that classical communication involves assumptions and pitfalls and that we have to be very careful whenever we talk about it and what we're going to do today is standard practice in physics we're simply going to stipulate that these components A1 and A2 the separable components of Alice communicate classically and so we're going to draw out like this we're going to draw a classical Channel between them and that classical Channel passes through Bob it uses degrees of freedom of the ambient Photon field or the ambient atmosphere or pieces of paper or email or something like that that's out there in the environment to exchange classical messages and there are a couple of things to note here uh I haven't even written down here we're assuming that A1 and A2 share some language this classical communication that's a big assumption as we talked about last time um but I will note explicitly that classical communication takes time that's what defines classicality here it's causal so it can't work faster than the speed of light uh

this is a topological Theory we don't have any vetting geometry so we can't at this point talk about light which is a relationship between space and time because we've got time we've got clocks but we don't have any space but we can say that whatever passes through this classical Channel requires at least one clock tick on Bob's side so it's a process that Bob has to execute and executing that process takes time so getting uh the signal from me to you on the internet takes some time so the environment's clock has to tick during that and all of these assumptions bundle together turn out to be a fairly serious set of assumptions and they involve assumptions about fine-tuning about the universe being just right and uh they should make us nervous and I just want to leave it at that we always assume with complete a plum that agents can communicate classically but it's not a straightforward assumption and it's worth any of you are looking at for interesting uh research projects can look into the nature of classical communication it's full of hidden assumptions and surprises and things that aren't at all straightforward even though they look like they ought to be so there's a challenge so why do we Ascend this why do we assume class communication we do it because we have to if we're going to talk about detecting entanglement most of you probably know the history of of the notion of entanglement it was first noticed by Einstein podolsky and Rosen uh Ever After called epr in 1935 and they wrote a paper which is Infamous uh and called the epr paper in which they argued that the phenomenon of entanglement made no sense and couldn't possibly be physical and since quantum theory included this Con this this notion of entanglement uh it could possibly be a complete theory of what's going on in the world and Schrodinger uh replied the same year with a very detailed paper where it introduces among other things Schrodinger's cat as a paradox and he explains very clearly what entanglement is and why entanglement follows from quantum theory and he uses that discussion to argue that the classical notion of a physical state is not actually applicable to the world as it is so he answers the epr paper very directly of this discussion of is currently most mostly known as the boar Einstein debate because of course Bohr already in 1928 has said had said in print that the classical notion of causality in space and time simply doesn't apply anymore and Einstein objected to that vigorously but that was 1935 and it was almost 50 years later that entanglement was first observed in the laboratory here the 1982 picture is a picture of the interior of Allen s that's Laboratory uh when he performed the first entanglement experiment for which uh he finally won the Nobel Prize last year and uh these days this kind of experiment is done all over the world and is even done with satellite-borne sources Ian resource is the key to Quantum security Quantum secured communication as we'll discuss in

Quantum cryptography and Quantum banking the Quantum Internet all this other stuff that you may have heard of so uh before talking more about these experiments I just want to uh provide a definition of entanglement uh entangled simply means not separable that's the definition so a state a joint State ρ is entangled if and only if it does not Factor into two separable states and that's it that's the definition the literature and especially the popular literature are is absolutely rife with hype and various kinds of misdirection of entanglement and and authors trying to convince you that entanglement is this amazing uh kind of woo-woo thing and whenever you run across that go back to the definition I mean entanglement simply means uh this simple mathematical condition of not separability or a state that isn't factorizable and that means that the state does not display classical conditional Independence so classically that's the way to think about it it's a failure of conditional Independence and it's it's ubiquitous in quantum theory even though it wasn't actually observed for almost 50 years after it was predicted so here's what all these experiments look like they're called Bell epr experiments after John Bell uh who derived the statistical Criterion for detecting entanglement and epr Einstein podsolsky and Rosen who did this as a thought experiment and said if this this occurs and the theory it can't possibly be real and of course it is real uh as now amply demonstrated by experimentation here's how they work there's a source The Source produces a state that's that is entangled and I've I've written the typical uh what's called a bell State it's $\frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle)$ just to normalize it but up plus or minus down is clearly not the same as up times down so this is not a separable state and this state propagates outward in space and time to two observers who have independent Laboratories uh for example on different continents in that space-based experiment that I showed you the picture of from the cover of science a few years ago 2017 I think the two observers have freedom to choose their own reference frame their own z-axis for measuring spin they do a spin measurement and they do this over and over and over again they accumulate statistics and they exchange their results so that they can analyze the joint statistics of their experiment and they check to see whether that joint statistics is consistent with classical probability and if it isn't uh to a statistically significant level then they've been able to detect entanglement though notice that this process requires they have to make their independent measurements in their own Laboratories and then they have to exchange results or one of them has to send results to the other one or they both have to send their results to some third party but in all of these cases they're exchanging some classical data and so we have the assumption that whoever gets that classical data knows the language it was recorded

in and so can do a joint analysis so I'll just show you another picture of this experiment that's in a standard space-time diagram here time is vertical space is horizontal of the two observers are labeled Alice and Bob they share a Quantum channel that goes from Alice to the source to Bob that's the entangled state they do some processing and then one of them here Bob sends the results to Alice and that's a classical communication it takes time and it traverses space so this step is classical so this is why we have to assume classical communication to talk about detecting or using entanglement so now I want to represent this in a different way uh using our little picture earlier we're going to think of Alice again as two components A1 and A2 they're separable so they have to communicate classically they have a classical channel that goes through Bob and they also now share some Quantum Channel they share an entangled state and the entangled state of course is produced by Bob the source is part of and their detectors if you want to think about it that way are located on this boundary B or they're interfaced with the sources on this boundary V and each of them executes a process that accumulates data from the quantum Channel and communicates it to the other Observer via the classical chat now this picture illustrates what's called a lock protocol lock means local operations and classical communication so A1 and A2 are each operating locally on their boundary they're making measurements locally on their boundary and they're communicating via actions locally on their boundary and observations locally on their boundary so for example I'm communicating with my computer here we can consider it my boundary and stuff is happening in the environment that's implemented by the internet and you're receiving those Communications locally on your boundary on your laptop or whatever it is you're using so that's what local operations means these protocols specifically involve classical communication and so they're able to detect entanglement in a bell epr experiment is just an example of a lock protocol uh The Observers exchange information about how they're going to do their experiments they're both going to measure D spend they're going to use independently chosen reference frames and they're going to look at the output from some agreed upon source so that's all classically set up then they they actually make their observations and then they classically communicate their results so classical communication front and back lot Protocols are not limited to this sort of of Bell epr sort of setting we can also represent a scattering experiment as a lock protocol here's a representation of scattering as a lock protocol and you can see all I've done is uh cut that previous boundary at the division between 81 and A2 and I've then extended it in some external time some um the time that is counted by Bob's clock not by Alice's that's what's meant by external and I've relabeled Alice II as Alice 1 at a later

time so now what Alice is doing is communicating with her past self or A1 is communicating with her future and they're each doing something different uh A1 is executing a procedure that in the laboratory we call State preparation she's preparing a state that we'll call in which is just the initial state of some scattering experiment so for example at the LHC uh in is some state of two protons that are quartering around the ring and tlhc at close to the speed of light and uh Alice then later on at $t + \Delta T$ is doing a different kind of quantum process she's uh analyzing the data coming out of something like the atlas detector which is a multi-particle uh counter that's located at the LHC so they share a Quantum Channel which is the scattering channel that takes the incoming State and produces the outgoing state and I'll recall from the very first session as Feynman told us to understand this transition from in to out we have to integrate over all possible paths that are consistent with conservation of spin helicity basically momentum plus spin all the different spins that are relevant in a high energy physics experiment including things like lepton number which is basically a spin now how do we know how do we know what we have to conserve we know what we have to conserve because Alice is doing the experiment has a classical memory that allows her to remember what she prepared she can remember that what she did you know back in the past was set to protons counter rotating in the LHC and they had certain energies and so they had certain momentum and they were protons so they had certain spin variables uh all of that has to be remembered to make any sense of the output of the detectors and so to actually come up with a measured state that makes sense and very often this this memory channel is neglected by physicists and I'm going to pause here and let Andrew ask a question uh thank you Chris uh so does this imply that every time you have a log channel the quantum part is a like a scattering picture like a unitary Evolution even in the earlier epr picture is that Quantum Channel portion always a unitary Evolution scattering picture thank you um yes for essentially the reason that uh Define the class classical Channel you're always assuming you have something that's protected from decoherence by the environment so uh yeah if you're doing an epr experiment you have to assume that you don't have any environmental decoherence between the source and the detectors and similarly when you're doing a scattering experiment uh you have to make the assumption that the environment is not reaching in there and messing with uh the outgoing state that you're going to see or reaching in and messing with the incoming State and changing it in some way that re-prepares it uh in a way that you don't know that you have no knowledge of so yes you're making a fairly strong anti-decoherence assumption in any of these settings when you're talking about having a Quantum Channel does that does that answer the

question uh yeah thank you okay great so we can think of we can think of lock we can think of scattering as lock and I I just wanted to now emphasize that effectively what Locke is implementing is two parallel memories a which as we said in the case of scattering is memory for Preparation the pr case it's memory for preparation for what the experimental setup is for what the instructions for doing the experiment are and we also have a Quantum Channel which is effectively another memory it's something that's extended Through Time and again it's implemented by the environment implemented by Bob and it involves Bob's clock ticking whenever we have two memories in parallel we can construct an error correcting code so think of a simple classical error correcting code for example the use of parity bits uh in exchanging Digital Data so if I send you a bit string classically and I send it to you two or three times uh so there's some redundancy in my signal and I send you a parity bit I say Okay add up your bit string and the result should be one it's one on my side uh if you've added up and you get zero there's an error someplace you can use that parody bit to detect errors so you can throw out the the bit strings that you receive that have parity zero and say oh the environment uh interfered somehow with the signal or some eavesdropper interfered with the signal that made a mistake or something like that so this corresponds uh to environmental decoherence not being a good assumption even though it's classical environmental interference is not a good assumption in this case so a Quantum error correcting code is basically the same idea I use some feature of the fact that I have two different memories and that uh I'm encoding something in time I'm encoding multiple replicants of something in time to provide a way of checking to make sure that I'm sending the correct message so uh I can for example consider the message to be my classical memory and consider the quantum process as a process that tests that memory so I remember that I need to do experiment X and I will get result y I actually do experiment X on this CL on this Quantum Channel and see if I get result y uh and if I don't get result y then something's wrong either the either I can't trust the quantum Channel uh or I've I've remembered my my preparation conditions incorrectly so I'm using one memory to check the accuracy of the other and this Pro of using something about the environment some channel in the environment to check is absolutely ubiquitous and uh we as it all the time so for example look around your room and your room hopefully looks more or less the same now than it did at the start of this session and that's very reassuring you know it it reassures you that something's not happening in the environment that's dangerous and disturbing and if you couldn't do that if if your environment was constantly changing and there is no familiarity about the environment you wouldn't be able to check your memory uh you wouldn't be able to

check your sanity so even though we don't think about our local environments as effectively an error correcting code that's exactly how we use it we use stigmergic memory or memory out there in the environment as a way of checking our own memories and assuring that our memory is okay so uh I talked a little bit about Quantum security uh here's how Quantum security system works again it's a lock protocol uh we have some classical signals that can be thought of as a code or a key or a one-time pad for example that encodes a way of doing experiments and once that's been exchanged assuming it's secure then we can use the CL the quantum channel to communicate information and the classical channel has told us what we have to do with that Quantum Channel we have to make certain kinds of measurements at say particular clock times as measured and we'll get information sent from from the past from the sender the classical Channel protects the quantum channel from adversarial agents because anything that an adversarial agent does to the quantum Channel if if some agent tries to eavesdrop for example on the quantum Channel they'll disrupt the quantum Channel and the results of our measurement uh will not be what's expected given the protocol that we run that we have run Quantum communication is secured by the fact that interfering with some entangled State actually destroys the state so it's detectable we can always know if someone is trying to disrupt our Communications which isn't true if we just have two classical channels Charlie can always intercept a message and then recreate it and that's not possible with a Quantum Channel so let's refold this diagram uh so that we've taken the external time back out of it and we have this picture where the two components of Alice are just interacting via their shared boundary with their environment Bob and now we can see that the source of redundancy is not time it's the boundary itself just sharing the boundary allows uh Alice's components A1 and A2 to put copies of the same information on different places on the boundary so the boundary now becomes a resource for redundant storage of information so Alice wanted Alice 2 can use different parts of the boundary to check what other parts of the boundary say to check the information that's encoded on other parts of the so they can define an error correcting code on the boundary itself using this I just want to leave you with this idea using classical and quantum channels together allows us to Define error correcting codes on boundaries that we share as separate agents leave you to think about what shared measurements what shared Quantum reference frames let us as agents see the same things and this is what we'll talk about next time in September we'll talk specifically about how space provides us with an error correcting code and try to understand why space is treated as emergent from something else basically from communication in in most of quantum gravity so

thank you very much uh I encourage you to use the interactive QA and to attend the discussion sessions and we'll see you in September thank you awesome Chris thank you for the lecture um Andrew do you have any remarks or questions no not right now thank you I'll just ask a few quick questions and if anyone has one quickly in the live chat so you mentioned the tick of the clock and in active inference we often hold up side by side both the discrete time models and the continuous time models so do clocks have to tick discretely or how do we think about continuous time modeling uh in this in this Quantum setting they do have to take discretely because one has to be able to distinguish uh Alice's preparation of the qubits on her boundary from Bob's measurements of the qubits on the boundary and uh this process can be extremely fast with respect to any kind of sense of macroscopic time oh you know your your eyes are responding to photons in I think it's hundreds of femtoseconds so the the discrete transitions that are effectively time measurements for you are at that very fast molecular time scale much much faster than our typical clocks now when we when we try to do very fast timing we're constructing a faster clock than that but it's a clock that we can only observe indirectly awesome okay now this one maybe connects a little bit to application but with all the developments happening in theory some of which you're reviewing here and also in practice with Quantum computation and then keeping in mind this distinction sorry I've lost you back I've Lost You Daniel okay we'll wait till you're back I can see that no okay okay yeah all right just repeat all right motion please um yeah so so given these um rapid and ongoing developments in theory and practice related to Quantum information science and and Quantum information security how are people using this distinction of the classical stigmergic memory and then the quantum cognitive process to anticipate how to develop information systems that are reliable going forward I don't I don't know of any instances in which of thinking about Quantum cognition has entered directly into any design of communication communication systems oh the ability of humans to interact with these systems is just assumed so it's just assumed that the user can interact with say a computer interface which is the sort of interface that will be provided or is provided to any quantum computer so if you think about a quantum computer a practical quantum computer it has Classical interfaces front and so in thinking about interacting with the computer I'm giving it some classical problem statement for example and asking for some classical answer and what it's going to give me is a probability distribution of classical answers but I can by running the calculation many times make that probability distribution fairly tight so so yeah I don't know if anyone talking about Quantum bcis at this point it'd be an interesting question um and was it always understood these sorts of

blind spots and axioms associated with classical communication was it was it developed as kind of like a proposal or like a special case and and uh you know more unpacking was left for later or are these blind spots things that were now coming into awareness of given what we're learning in the quantum information Sciences I think we're learning how serious they are uh but they were certainly recognized even by Bohr in his 1928 paper um that's a very good paper too it's it's called Uh the present state of quantum theory or something like that and it's in nature uh and it's available on the web somewhere uh I I have a copy of it uh but it was written uh in a sense it's Bohr's summary of the 1927 solve a conference which was the conference at which um everyone doing quantum theory at that time got together to talk about their results and in it uh Bohr essentially presents what came to be the Copenhagen interpretation which I think is probably unfairly called an interpretation it's sort of the the Copenhagen pragmatic stance toward quantum theory and essentially what Bohr says is uh that if we're going to do science we have to assume that people can talk to each other and if we're going to do science we have to assume that there are instruments that we can manipulate and have reasonable confidence that we know what manipulation we've done so if if I turn a knob from one to two I have to be reasonably confident that I'm making a particular change that I can understand and we have to uh have reasonable confidence that different Laboratories can replicate the same experiment and if we think about any of those assumptions in purely Quantum theoretic terms then all sorts of questions are raised right every Quantum system is unique we have the no cloning theorem so every scientific instrument is unique so there's no such thing as as exactly replicating an experiment right that's that's a coarse grained description of what I've done with a completely unique Quantum system and my brain of course is unique uh and yours is too and everyone else's so talking about a sharing a language is essentially a coarse grained assumption and what the Copenhagen stance is all about is coarse graining yielding effective classicality or what that really means is coarse graining yielding a description that we can use in in our in the language that we understand so these are very very old issues that have been recognized for a long but I think in part because of the disaster of shut up and calculate uh they haven't really been talked about in the classroom awesome I'll just read short questions from the chat we can just uh we can pass them to the discussion section but but there's a lot of great diverse questions so Alexi wrote I think I've Lost You Again the Revenge of Zoom well we looked at the chart earlier where like Bob was talking to Alice the unidirectional communication okay good yeah okay okay okay the unidirectional the third party is recording it I'm sure um Alexi wrote uh regarding the environment for uh being used for error correction

a symptom of derealization in Psychology psychological trauma breakdown we don't recognize the environment around us and it's scary makes sense makes sense all right from Bala nice presentation I heard Chris say entangled states are local for a specific reference frame is this entanglement unitary or is it related to the numbers of degrees of freedom I think the the quick answer would be uh if you have a complex system with many degrees of freedom some can be entangled While others aren't uh it depends on what parts of the state are separable so again entanglement is just a statement about uh components components of a joint state that factor or don't so you can have a joint state that looks that looks separable along some Dimensions but does know the dimensions another big complication here is that whether a state looks entangled depends on the reference frames that you use to measure it with and if you if you pick entangled reference frames then what we ordinarily think of as separable as entangled States become separable so you can move entanglement around awesome okay from Roland Rodriguez is the discrete temporality only within the context of a given holographic screen interface with time being bracketed otherwise Beyond interface systems is continuity or discretion kept agnostic I think if I understand the question correctly it's yes uh if I don't draw any boundaries I just have the joint state then the evolution is unitary and uh in a sense time plays no role and if if unitary evolution is perfectly time symmetric so time actually drops out altogether awesome so talk about time we have to draw boundaries awesome time must have a stop Huxley and I'll ask one last question from the appropriately monitored question quanderer all communication is abductive is a sentiment that I've been using to develop some arguments does that scan how do we connect what we're discussing here with Quantum to abductive logic yeah I I I could go along with that I'd want a lot more detail uh all communication is is essentially analogical I I would agree with that yes enter yes thank you um uh so going back to the picture where the scattering picture um so we have so far everything is is topological and um we certainly induces you know breaking the boundary in different sectors I suppose it induces a topology on the boundary now I had a couple questions as for the geometry and this might be getting ahead is it is it an emerging property of those channels between different uh boundary sectors and if if that's the case then what what does it mean to have a qrf for space um yeah this is this is this is really the topic for September but uh you you do raise you do point out the fact that it's something of a chicken and egg situation uh uh the way I think of it is I suppose to choose the chicken side and think about uh what kind of reference frame does the agent have to implement to to see space on their boundary and the the reason I think about it that way is that's a question I want to be able to ask about biological systems and and this

actually gets into probably what won't be talked about until October but uh I I want to be able to ask essentially a phylogenetic question of what sorts of organisms where and the lineage from Luca do uh we find systems that are able to assign spatial coordinates to their boundaries and what coordinates come first and I I suspect that that sort of discrete location names with no metric come first that A system can talk about for example uh this part of their membrane versus that part of their membrane without even having any ability to say how far apart they are and the that some sort of metric representation of this angular space comes later and that a radial space comes even later than that but that's that's at this point speculation but I think these are questions that we can end up asking uh given the right kind of formalism and the right understanding of what it means to to to be able to make a spatial so the conjecture is that uh the metric would emerge from a discrete Network metric of counting ticks sort of thing yeah yeah I think that's I think that's correct and uh that the sort of the sort of symmetries of space that we see are very much tied up with object identity so if we think about um you know something like rotational symmetry let's see uh I should stop screen sharing so you know I've got this object right and I can do this and I say oh you know space has this rotational symmetry but for me to say that I have to recognize that this is the same object right I I have to implement object persistence which we all learn to do when we're nine months old or something but uh we tend to regard those as separate things as you know we we learn object persistence because space is rotationally symmetric but we can view it the other way around that space is rotationally symmetric for me because I Implement these constraints that are called object persistence okay do you think we can understand phylogenetics as a whole as a kind of scattering experiment like the high energy leuka event and then dissipating and propagating in a very Tangled Bank well I I I do want to be able to understand phylogeny in exactly the same terms as as embryology Mike Levin and I have published papers about that that we should be able to view the lineage of Luca in exactly the same way that you know we we view the lineage of the zygote that produced us so that's why it's it's no coincidence during pregnancy uh the the development of the embryo merus evolution in terms of the parts that are developed first yeah this is the old idea that that phylogen and you're not an endogeny are are related in some way which from an embryonic development point of view um they're certainly analogous in ways and uh sorry if I may ask one last question that I think should be very quick uh regarding that the weak rotation I'm still a bit unsure of how it ties to to this uh diagrammatic notation that we had uh with Quantum channels I'm sure so on in the second session we briefly talked about the weak rotation um you know being a a reversal in the

direction of time is there any relation to today's session yeah I whenever we think about uh the flow of time as driven by the input output cycle the the input output cycle for agents separated by a boundary is is always there's always a sign change right my input is your output and vice versa when we think about the Bob's clock ticking when Alice does his direction of direction of time in a sense points with respect to the boundary opposite to hers but that's only with respect to the boundary he sees he sees information flowing in and she sees information flowing in that's that's the only point being made here about the wick rotation as an operation right thank you thank you Chris very well appreciated by the viewers and by us so we'll certainly look forward to reviewing the Q a and everybody is welcome and encouraged to join for the participatory discussion in about two weeks on Saturday so till next time all right thank you cheers